Citi Bike coverage and neighborhood mobility

Timothy Phan and Ethan Deng · Group PAIR · Midterm progress report

Research question and current state

How did observed Citi Bike station density and usage vary with median household income and the share of households without a vehicle across New York City census tracts in September 2025, citywide and within each borough? The geography, month, demographic variables, and descriptive approach remain as proposed. The audience is NYC transportation planners and community groups assessing where observed coverage merits closer review.

Completed: full-month acquisition, documented cleaning, point-to-tract spatial join, citywide/borough comparisons, one map and five charts, and a reproducible Python workflow. Static figures are the midterm milestone. Animated maps, the webpage, and operational/boundary sensitivity checks remain final-project work; no historical station inventory has been replaced by a current feed.

Data and documented cleaning

All six original trip CSVs contain 5,287,447 rows. ACS 2020–2024 five-year tables B19013 (income), B08201 (households/vehicle availability), and B01003 (population) supply tract context; 2024 TIGER/Line supplies boundaries and land area. ACS data were obtained from official table-based text files without an API key. Raw files, download dates, URLs, and SHA-256 checksums are retained.

The sequential trip filters remove 931 starts outside September; missing ride IDs, duplicate IDs, and invalid start timestamps each affect 0, 0, and 0 rows, respectively. Of 5,286,516 retained starts, 2,118 lack both a start station ID and usable coordinates and cannot enter station usage. Missing destination IDs affect 12,865 retained trips but do not remove their valid departures.

Stage	Before	Affected / excluded	After
Missing ride ID	5,287,447	0	5,287,447
Duplicate ride ID	5,287,447	0	5,287,447
Invalid start timestamp	5,287,447	0	5,287,447
Start outside September	5,287,447	931	5,286,516
Eligible residential tracts	2,327	96	2,231

The spatial join assigns 2,212 of 2,269 located station IDs to NYC; 57 station locations outside/unmatched to NYC remain audited. There are 0 boundary ties. Residential comparisons retain 2,231 of 2,327 NYC tracts; 96 fail the positive-household/population/land-area rule. Another 32 eligible tracts lack income and remain mapped but are excluded only from income comparisons. The 13 upper-bound income estimates are flagged; 272 eligible tracts have income MOEs above half their estimate.

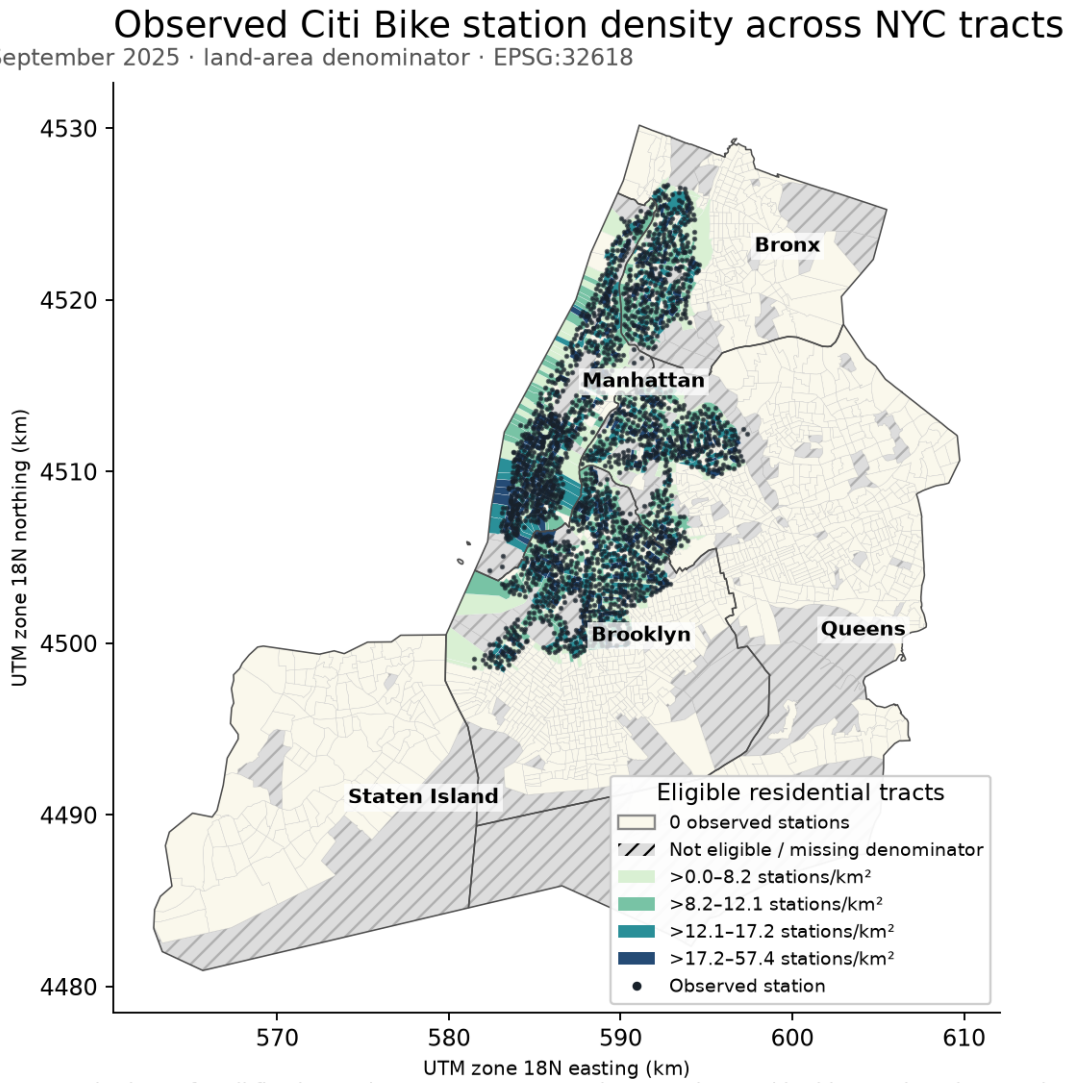
Methods and validation

1. Parse local timestamps, keep September starts, and deduplicate ride IDs across files using a disk-backed index. 2. Infer station locations from both endpoints by modal coordinate pair, preserving string IDs. 3. Project to **EPSG:32618 (meters)**, spatially intersect stations with tracts, and merge ACS by eleven-digit GEOID. 4. Aggregate starts by their station's tract; calculate stations/land km², stations/10,000 residents, and departures/observed station. 5. Compare unweighted tracts using fixed citywide income and no-vehicle-share quartiles within every borough; report descriptive Spearman correlations. Coverage includes zeros; usage excludes zero-station tracts. Land area is TIGER ALAND/1,000,000, excluding water from the denominator.

All 15 reconciliation checks pass. The 5,284,398 assigned NYC departures reconcile across stations, tracts, boroughs, and dates; 196,150 are in ineligible residential-comparison tracts. The notebook regenerates outputs from raw inputs using relative paths.

Early results: coverage across all five boroughs

Observed coverage is uneven: 1,335 of 2,231 eligible tracts (59.8%) have no observed station. The share is 11.6% in Manhattan, 72.9% in Queens, and 100% in Staten Island. These are observations from trip records, not a claim that residents cannot reach stations in adjacent tracts.



Station coverage is shown for all five boroughs; zero means no station was observed in this month's trip records.

Sources: Citi Bike, September 2025; ACS 2020–2024; TIGER/Line 2024. Group PAIR. Positive-density classes: citywide quartiles.

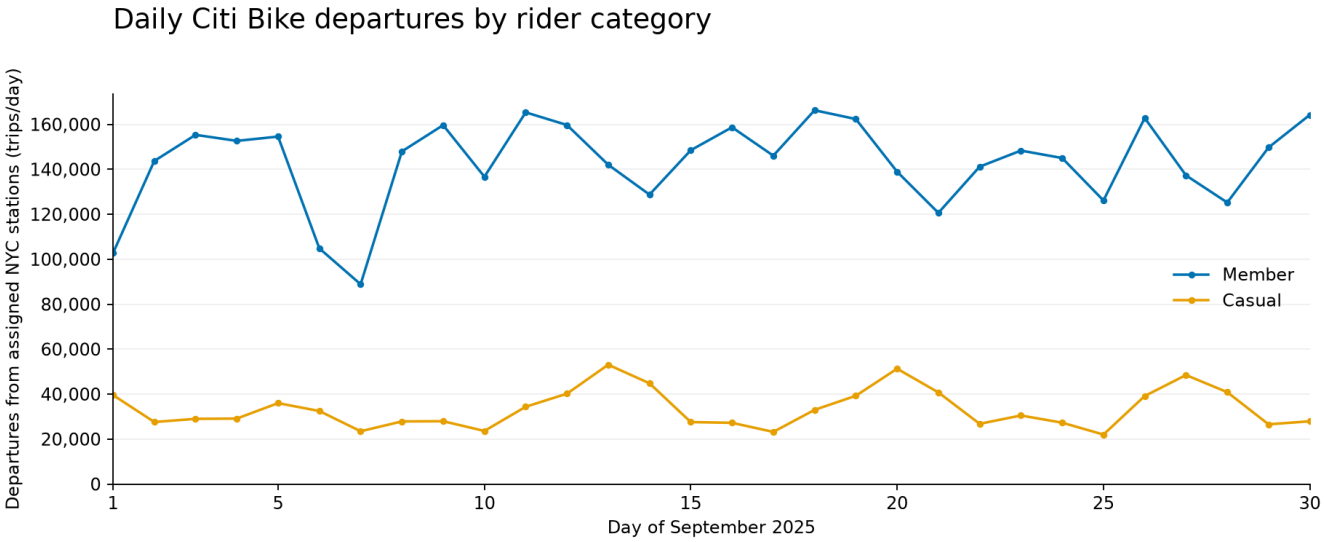
Borough	Eligible tracts	Zero stations (%)	Median stations / km²	Sept. departures*
Bronx	344	58.4	0.00	139,926
Brooklyn	777	61.3	0.00	1,446,013
Manhattan	301	11.6	11.55	3,342,360
Queens	687	72.9	0.00	356,099
Staten Island	122	100.0	0.00	0

*Departures include all assigned NYC stations, including tracts outside the residential comparison. Density medians and zero percentages use eligible residential tracts only. Figure 01 uses distinct zero/missing classes; full-size PNGs accompany this report.

Early results, limitations, and plan to finish

The citywide income–density correlation is nearly flat (ρ=0.05), but borough patterns differ: Bronx -0.53, Brooklyn 0.30, Manhattan 0.05, and Queens -0.13. No-vehicle household share is more positively associated with density citywide (ρ=0.59). Its highest quartile has 21.9% zero-station tracts, versus 95.5% in its lowest quartile. The borough differences caution against a single citywide income story.

Among tracts with stations, median September departures per observed station rise from 496 in the lowest income quartile to 4,068 in the highest. This conditional comparison describes activity where stations were observed, not demand in unserved tracts or trips made by residents. Daily variation is visible below; weather, weekends, and holidays have not been modeled.



Daily departures show the full month's observed activity by rider category, including stations in nonresidential NYC tracts.

Sources: Citi Bike, September 2025; ACS 2020–2024; TIGER/Line 2024. Group PAIR.

Interpretation and remaining risks

ACS estimates cover five years and have sampling uncertainty; they do not describe individual riders. Tract boundaries, jobs, transit, land use, and partial-month operation can affect the patterns. The one flagged station location spans 291 m; its modal pair accounts for 383 of 389 endpoint observations, and both locations fall in the same tract. We retain the modal location with no manual override. Low observed density is not a direct walking-access measure, and low usage does not establish low or unmet demand.

Plan to finish and requested feedback

Ethan · Nov 16: review land/population normalization and boundary effects; repeat usage comparisons for stations observed in both first and last weeks. **Tim · Dec 6:** build the two interactive maps, daily animations, fixed scales, borough filters, and GitHub Pages draft. **Both · Dec 15:** review interpretation, rerun the workflow, and finalize. Main risks are uncertain exposure, ACS precision, and map performance; use sensitivity checks and aggregated layers rather than adding causal claims or rendering individual trips.

Feedback requested: (1) How should nearby stations across tract boundaries be represented? (2) Are these descriptive comparisons sufficient given ACS uncertainty and correlated land use? (3) Is first/last-week station presence a useful check for partial-month exposure?

Companion figures: 02 income-density distributions; 03 vehicle-share-density distributions; 04 usage by demographic quartile; 05 demographic scatterplots. Full citywide/borough statistics and uncertainty flags are included with the notebook.

Sources and code: Citi Bike trip histories; ACS 2024 five-year Summary File; TIGER/Line 2024. github.com/EthanWD1/cp101-citibike-equity. Full provenance, attribution, and computational-assistance notes are in the README.